Grade 11 - Term 3
C1–C4 Practice Exercises and Solutions

Contents

Evaluated criteria

- **C1** Characterises continuous random variables by distinguishing them from discrete variables and interpreting their probability density function, cumulative distribution function, and support in economic contexts.
- **C2** Applies uniform and triangular continuous distributions with constant or piecewise-linear probability density functions to calculate interval probabilities using geometric areas.
- **C3** Applies circular-arc probability density functions to calculate interval probabilities using the geometry of circles and circular segments.
- **C4** Applies the exponential distribution to constant-rate waiting-time situations and assesses its suitability.

1 Identifying Probability Distributions from Contexts [C1-5]

A variety of Colombian organizations use probability models to represent uncertain quantities and events. In each situation below, a random variable is defined from the information provided. The appropriate probability distribution must be identified from the characteristics of the situation.

C1 For each context, characterize the random variable and identify the appropriate probability distribution. **[5]**

Context 1. A payment platform monitors one scheduled electronic transfer. By the end of processing, the transfer is either completed successfully or not completed. The platform uses this information to estimate verification and support needs.

Context 2. An agricultural technology company monitors the daily temperature variation inside a refrigerated storage unit. The variation can take any value from $-2°C$ to $2°C$, with all values in this interval considered equally likely. The company uses this model to set an operating margin.

Context 3. A streaming platform sends the same offer to exactly 80 randomly selected eligible users. Each user either subscribes or does not. The platform records the number of users who subscribe to estimate the server capacity needed for the promotion.

Context 4. A Bogotá parking company records the number of vehicle breakdowns during a fixed one-week period. Comparable records show that breakdowns occur independently and without a predictable timetable. The company uses these records to help schedule maintenance staff.

Context 5. An online marketplace studies the delivery time for a particular type of order. Delivery times range from 20 to 80 minutes, with deliveries around 45 minutes occurring most frequently. They become progressively less common toward either extreme. The company uses this model to estimate delivery times.

Context 6. A quality-control laboratory tests electronic components until the first defective component is found. Components are tested independently, and testing stops after the first defect. The laboratory records the total number of components tested, including the defective one, to estimate inspection costs.

Context 7. A cloud-computing provider studies storage usage among small business customers. Usage ranges from 0 to 50 gigabytes and becomes steadily more common toward the upper limit, without reaching a peak before 50 GB. The provider uses this model to plan capacity.

Context 8. A retailer sends a promotional message to exactly 120 customers selected using the same procedure. Each customer either makes a purchase or does not. The retailer records the number of purchases made in response to the message to estimate the inventory needed for the campaign.

Context 9. A coffee exporter measures the moisture content of coffee bags under usual storage conditions. Measurements cluster around 11%, with approximately symmetric behaviour around this value. Values become less common farther from 11%. The exporter uses this model to establish quality-control thresholds.

Context 10. A food-delivery company records the number of customer cancellations during a fixed three-hour dinner period in one delivery zone. Comparable evenings show cancellations occurring independently without a recurring timetable. The company uses these records to estimate customer-service needs.

C1 - Distribution identification: Solutions **[5]**

1. **Context 1 — Discrete, Bernoulli.** X_1 records whether the transfer is completed or not.
2. **Context 2 — Continuous, Uniform.** X_2 takes any value in $[-2, 2]$, all equally likely.
3. **Context 3 — Discrete, Binomial.** X_3 counts subscribers among 80 users, each with two outcomes.
4. **Context 4 — Discrete, Poisson.** X_4 counts independent breakdowns during a fixed period.
5. **Context 5 — Continuous, Triangular.** X_5 has a minimum of 20, maximum of 80, and mode of 45 minutes.
6. **Context 6 — Discrete, Geometric.** X_6 counts tests until the first defective component.
7. **Context 7 — Continuous, Linear.** X_7 has a density that increases steadily toward 50 GB.

8. **Context 8 — Discrete, Binomial.** X_8 counts purchases among 120 customers, each with two outcomes.
9. **Context 9 — Continuous, Normal.** X_9 is approximately symmetric and concentrated around 11%.
10. **Context 10 — Discrete, Poisson.** X_{10} counts independent cancellations during a fixed period

2 Delivery Time for a Local Grocery Order [C2-3]

A Colombian grocery-delivery company studies the additional time required to deliver orders during a busy evening period. For a particular delivery zone, the added delivery time X can range from 0 to 10 minutes. A delay of any length within this interval is possible, and shorter delays become steadily more common. Historical data indicate that no additional delay is 4 times as common as a 10-minute delay.

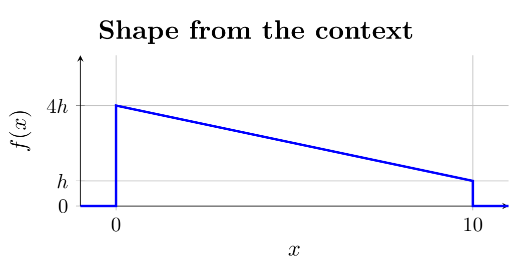
Questions/Tasks.

C2 Determine the probability that the added delivery time is between 3 and 7 minutes. **[3]**

C2 - Linear probability distribution [3]

2.1 Support and Shape

The random variable X has support $[0, 10]$. The density decreases linearly, and its height at 0 is four times its height at 10. Let the endpoint heights be $4h$ and h.



2.2 Total Area = 1

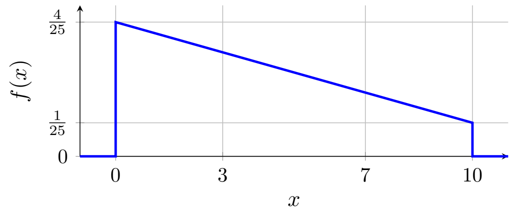
The total area under a probability density function is 1:

$$1 = \frac{1}{2}(10)(4h + h) = 25h, \qquad h = \frac{1}{25}.$$

Therefore,

$$f_X(0) = \frac{4}{25}, \qquad f_X(10) = \frac{1}{25}.$$

Completed density graph



2.3 Complete pdf

The density is the straight line through

$$\left(0, \frac{4}{25}\right) \qquad \text{and} \qquad \left(10, \frac{1}{25}\right).$$

Its slope is

$$m = \frac{\frac{1}{25} - \frac{4}{25}}{10} = -\frac{3}{250}.$$

Thus,

$$f_X(x) = \begin{cases} \dfrac{40 - 3x}{250}, & 0 \le x \le 10, \\ 0, & \text{otherwise.} \end{cases}$$

2.4 Probability Between 3 and 7 Minutes

The required probability is the area under the density from 3 to 7:

$$P(3 \le X \le 7) = \frac{1}{2}(7-3)\left(f_X(3) + f_X(7)\right).$$

Since

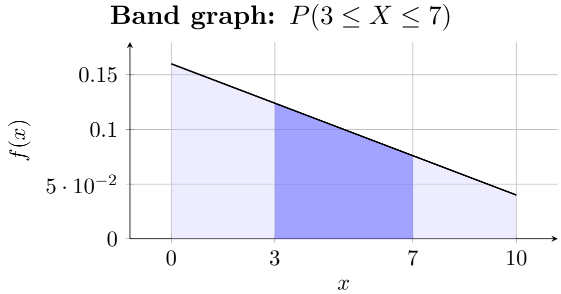
$$f_X(3) = \frac{31}{250}, \qquad f_X(7) = \frac{19}{250},$$

$$P(3 \le X \le 7) = \frac{1}{2}(4)\left(\frac{31}{250} + \frac{19}{250}\right) = \frac{2}{5}.$$

Therefore,

$$\boxed{P(3 \le X \le 7) = 0.40}.$$

Band graph: $P(3 \le X \le 7)$



3 Daily Commute Time for a Bogotá Apprentice [C2-4]

An apprentice records the time spent travelling to a training programme each day. Commute times range from 1 to 9 hours. Times become steadily more common from 1 hour up to a most common time of 4 hours, then steadily less common through 9 hours; there are no commute times right at either limit.

Questions/Tasks.

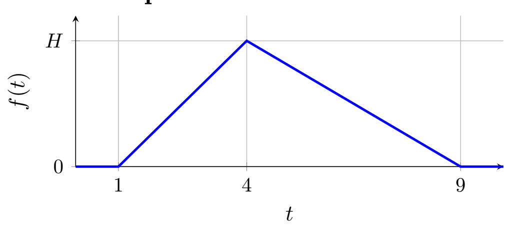
C2 Determine the probability that the commute takes between 3 and 7 hours. **[4]**

C2 - Triangular probability distribution [4]

3.1 Support and Shape

The random variable T has support $[1, 9]$. Its density rises linearly to a peak at $t = 4$, then falls linearly to zero at $t = 9$. Let the peak height be H.

Shape from the context

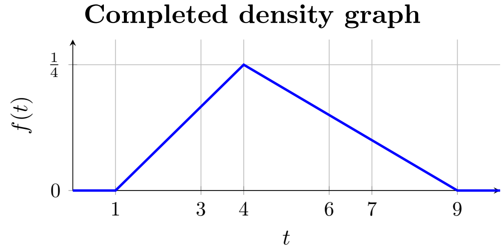


3.2 Total Area = 1

The triangle has total area 1:

$$1 = \frac{1}{2}(9-1)H = 4H, \qquad H = \frac{1}{4}.$$

Completed density graph



3.3 Probability Density Function

The rising segment joins $(1, 0)$ and $(4, \frac{1}{4})$:

$$m_1 = \frac{\frac{1}{4} - 0}{4 - 1} = \frac{1}{12}, \qquad f_T(t) = \frac{t-1}{12}.$$

The falling segment joins $(4, \frac{1}{4})$ and $(9, 0)$:

$$m_2 = \frac{0 - \frac{1}{4}}{9 - 4} = -\frac{1}{20}, \qquad f_T(t) = \frac{9-t}{20}.$$

Therefore,

$$f_T(t) = \begin{cases} \dfrac{t-1}{12}, & 1 \le t \le 4, \\ \dfrac{9-t}{20}, & 4 < t \le 9, \\ 0, & \text{otherwise.} \end{cases}$$

3.4 Probability Between 3 and 7 Hours

Use the complement of the two triangular tails:

$$P(3 \le T \le 7) = 1 - P(T < 3) - P(T > 7).$$

At the two relevant points,

$$f_T(3) = \frac{3-1}{12} = \frac{1}{6}, \qquad f_T(7) = \frac{9-7}{20} = \frac{1}{10}.$$

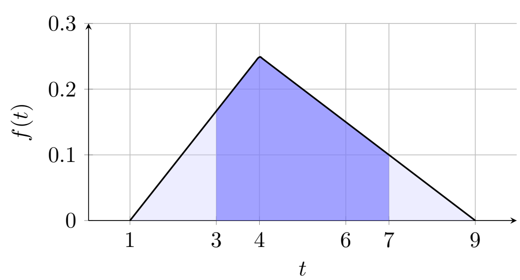
Thus,

$$\begin{aligned} P(3 \le T \le 7) &= 1 - \frac{1}{2}(3-1)\left(\frac{1}{6}\right) - \frac{1}{2}(9-7)\left(\frac{1}{10}\right) \\ &= 1 - \frac{1}{6} - \frac{1}{10} \\ &= \frac{11}{15}. \end{aligned}$$

Therefore,

$$\boxed{P(3 \le T \le 7) = \frac{11}{15} \approx 73.3\%}.$$

Band graph: $P(3 \le T \le 7)$



4 Timing of an Unexpected Market Movement [C3-2]

A financial monitoring system records the time of day at which an unexpected movement in an international currency market occurs. The movement is equally likely to occur at any time during a 24-hour period, so no part of the day is more likely than another.

Questions/Tasks.

C3 Determine the probability that the movement occurs between 9 p.m. and 5 a.m. **[2]**

C3 - Uniform probability distribution [2]

4.1 Uniform Distribution

Let T represent the time of the movement, measured in hours after midnight. Since every time during the 24-hour period is equally likely,

$$T \sim U(0, 24).$$

4.2 Probability Between 9 p.m. and 5 a.m.

The interval crosses midnight, so it consists of

$$21 \le T < 24 \qquad \text{and} \qquad 0 \le T \le 5.$$

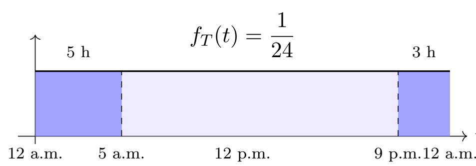
The total length is

$$(24 - 21) + (5 - 0) = 8 \text{ hours.}$$

Therefore,

$$P(\text{between 9 p.m. and 5 a.m.}) = \frac{8}{24} = \boxed{\frac{1}{3} \approx 33.33\%}.$$

Probability between 9 p.m. and 5 a.m.



5 Direction of an Unexpected Sensor Reading [C3-3]

A directional sensor records the direction in which it is pointing. The sensor can point in any direction around a complete circle, and every direction is equally likely. Let A represent the direction of the sensor, measured in radians from a fixed reference marker at 0 radians.

Questions/Tasks.

C3 Determine which is more probable and by how much: the sensor pointing within $\pi/12$ radians of the reference marker, or the sensor pointing within $10°$ of the reference marker. **[3]**

C3 - Uniform probability distribution [3]

5.1 Probability Within $\pi/12$ of the Reference Marker

The sensor can point uniformly around a complete circle of 2π radians. Being within $\pi/12$ radians of the reference marker includes the same angular distance on both sides of the marker, so the total angular width is

$$2\left(\frac{\pi}{12}\right) = \frac{\pi}{6}. \qquad \longrightarrow \qquad P\left(\text{within } \frac{\pi}{12} \text{ of the reference marker}\right) = \frac{\pi/6}{2\pi} = \boxed{\frac{1}{12} \approx 8.33\%}.$$

5.2 Probability Within 10° of the Reference Marker

A complete circle measures 360°. Being within 10° of the reference marker gives a total angular width of

$$2(10^\circ) = 20^\circ. \qquad \longrightarrow \qquad P(\text{within } 10^\circ \text{ of the reference marker}) = \frac{20^\circ}{360^\circ} = \boxed{\frac{1}{18} \approx 5.56\%}.$$

5.3 Comparison

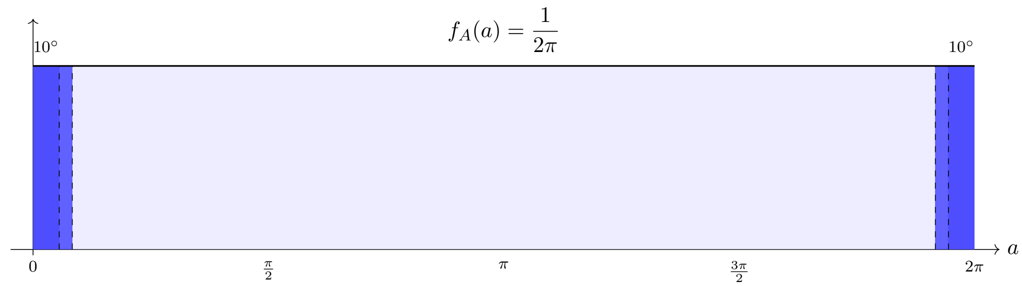
Since

$$\frac{1}{12} > \frac{1}{18}, \qquad \longrightarrow \qquad \frac{1}{12} - \frac{1}{18} = \frac{1}{36} \approx 2.78\%.$$

Thus,

$$\boxed{\text{Within } \frac{\pi}{12} \text{ radians is more probable by } \frac{1}{36} \approx 2.78 \text{ percentage points.}}$$

Comparison of the two regions



6 Waiting for a Study-Room Opening [C4-0.5]

At a university library in Bogotá, study rooms become available at an average rate of one room every 5 minutes.

Questions/Tasks.

C0.5 Determine the probability that the next study room becomes available more than 7 minutes after the start of the observation. **[0.5]**

C4 - Exponential probability distribution [0.5]

The average rate is one available room every 5 minutes, so

$$\lambda = \frac{1}{5} = 0.20.$$

For an exponential random variable,

$$P(T > a) = e^{-\lambda a}.$$

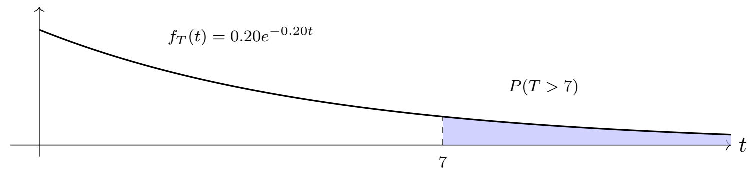
Thus,

$$P(T > 7) = e^{-(0.20)(7)} = e^{-1.4} \approx 0.2466.$$

Therefore,

$$\boxed{P(T > 7) \approx 24.66\%}.$$

Probability region



7 Waiting for a Market Price Alert **[C4-0.5]**

A financial monitoring system detects significant price changes in a commodity market at an average rate of 3 alerts every 20 minutes.

Questions/Tasks.

C0.5 Determine the probability that the next significant price change is detected within 6 minutes after the start of the observation. **[0.5]**

C4 - Exponential probability distribution **[0.5]**

The average rate is 3 significant price changes every 20 minutes, so

$$\lambda = \frac{3}{20} = 0.15.$$

For an exponential random variable,

$$P(T < a) = 1 - e^{-\lambda a}.$$

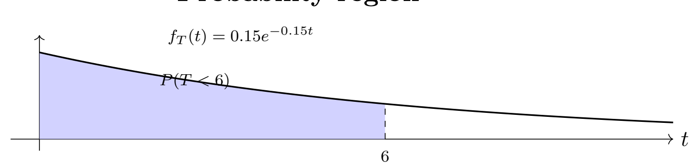
Thus,

$$P(T < 6) = 1 - e^{-(0.15)(6)} = 1 - e^{-0.9} \approx 0.5934.$$

Therefore,

$$\boxed{P(T < 6) \approx 59.34\%}.$$

Probability region



8 Waiting for a Demand Alert [C4-1]

An economic forecasting system detects significant changes in consumer demand at an average rate of 4 alerts every 30 minutes.

Questions/Tasks.

C1 Determine the probability that the next significant change in consumer demand is detected between 3 and 8 minutes after the observation time. **[1]**

C4 - Exponential probability distribution [1]

The average rate is 4 significant demand changes every 30 minutes, so

$$\lambda = \frac{4}{30} = \frac{2}{15}.$$

For an exponential random variable,

$$P(a < T < b) = e^{-\lambda a} - e^{-\lambda b}.$$

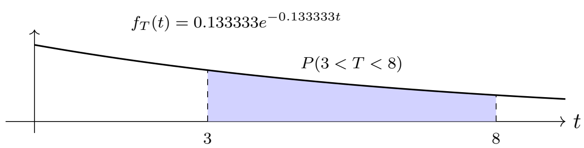
Thus,

$$P(3 < T < 8) = e^{-\left(\frac{2}{15}\right)(3)} - e^{-\left(\frac{2}{15}\right)(8)}$$
$$= e^{-0.4} - e^{-1.0667} \approx 0.3009.$$

Therefore,

$$\boxed{P(3 < T < 8) \approx 30.09\%}.$$

Probability region



9 Processing Investment Transactions [C4-1]

An investment platform processes buy and sell transactions throughout the trading day. The time between consecutive transactions is modeled by an exponential distribution. Based on previous observations, there is a 75% probability that the next transaction is processed within 8 minutes. On average, how many transactions are expected to be processed in a 10-minute interval?

Questions/Tasks.

C1 Determine the average number of transactions expected to be processed in a 10-minute interval. **[1]**

C4.9 - Expected number of transactions

There is a 75% probability that the next transaction is processed within 8 minutes. Using

$$P(T < t) = 1 - e^{-\lambda t},$$

we obtain

$$0.75 = 1 - e^{-8\lambda}$$

and therefore

$$\lambda = -\frac{\ln(0.25)}{8} \approx 0.1733 \text{ transactions/minute.}$$

The expected number of transactions in t minutes is

$$E[N(t)] = \lambda t.$$

Thus, for a 10-minute interval,

$$E[N(10)] = (0.1733)(10) \approx 1.733.$$

$$\boxed{1.733 \text{ transactions}}$$

10 Waiting for a Loan Approval [C4-1]

A financial institution processes loan applications throughout the day. The processing time for a randomly selected application is modeled by an exponential distribution. Based on previous observations, there is a 30% probability that an application takes more than 12 minutes to process. Find the average number of applications processed per minute.

Questions/Tasks.

C1 Determine the average number of applications processed per minute. **[1]**

C4.9 - Average number of applications per minute

There is a 30% probability that an application takes more than 12 minutes to process. Using

$$P(T > t) = e^{-\lambda t},$$

we obtain

$$0.30 = e^{-12\lambda}.$$

Therefore,

$$\lambda = -\frac{\ln(0.30)}{12} \approx 0.1004.$$

Thus, the average number of applications processed per minute is

$$\boxed{0.1004 \text{ applications/minute}}.$$

11 Comparing Investment Transaction Rates [C4-2]

Two investment platforms process buy and sell transactions throughout the trading day. The transactions processed by each platform are modeled by independent Poisson processes.

For Platform A, there is a 75% probability that the next transaction is processed within 8 minutes. For Platform B, there is a 60% probability that the next transaction is processed within 5 minutes.

Assume both platforms begin processing transactions at the same time. Using the average transaction rates, determine after how many minutes the expected difference between the numbers of transactions processed by the two platforms reaches 1 transaction.

Questions/Tasks.

C2 Determine after how many minutes the expected difference between the numbers of transactions processed by the two platforms reaches 1 transaction. **[2]**

C4.10 - Comparing transaction rates

For Platform A,

$$P(T < 8) = 0.75.$$

Using

$$P(T < t) = 1 - e^{-\lambda t},$$

we obtain

$$0.75 = 1 - e^{-8\lambda_A}.$$

Therefore,

$$\lambda_A = -\frac{\ln(0.25)}{8} \approx 0.1733 \text{ transactions/minute.}$$

For Platform B,

$$P(T < 5) = 0.60,$$

so

$$0.60 = 1 - e^{-5\lambda_B}.$$

Therefore,

$$\lambda_B = -\frac{\ln(0.40)}{5} \approx 0.1833 \text{ transactions/minute.}$$

Platform B has the higher average transaction rate. The difference between the rates is

$$\lambda_B - \lambda_A \approx 0.1833 - 0.1733 = 0.0100 \text{ transactions/minute.}$$

After t minutes, the expected numbers of transactions are

$$E[N_A(t)] = \lambda_A t, \qquad E[N_B(t)] = \lambda_B t.$$

Thus, the expected difference is

$$E[N_B(t) - N_A(t)] = (\lambda_B - \lambda_A)t.$$

We want this expected difference to reach 1 transaction:

$$(\lambda_B - \lambda_A)t = 1.$$

Therefore,

$$t = \frac{1}{\lambda_B - \lambda_A} = \frac{1}{0.0100} \approx 100 \text{ minutes}.$$

$$\boxed{\text{The expected difference reaches 1 transaction after approximately 100 minutes.}}$$